

Newman-Penrose Formalism Calculations

Stephani et al., Equation (23.8)

Metric

$$ds^2 = \left(-\sinh(t)^{-a+3} e^{(-8ax+4x)} \right) dt \otimes dt + \frac{\left(b^2 \cosh(t)^2 e^{(4ax+4x)} \sinh(t) + 24 e^{(4x)} \sinh(t)^3 \right) e^{(-8ax)}}{\sinh(t)^a} dx \otimes dx + \left(b \sinh(t)^{-a+1} \cosh(t) e^{(-2ax+6x)} \right) dx \otimes dz + \left(\sinh(t)^{a+1} e^{(4ax-12x)} \right) dy \otimes dy + \left(b \sinh(t)^{-a+1} \cosh(t) e^{(-2ax+6x)} \right) dz \otimes dx + \left(\sinh(t)^{-a+1} e^{(8x)} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$f = \sinh(t)^a e^{(2ax-10x)}$$

$$H = e^{(ax-x)} \sqrt{\sinh(t)}$$

$$G = \sqrt{\sinh(t)^{-a} \sinh(t)^3} e^{(-4ax+2x)}$$

$$B = b \cosh(t) e^{(-2ax-2x)}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \sqrt{\sinh(t)^{-a} \sinh(t)^3} e^{(-4ax+2x)} \right) dt + \left(\sqrt{6} \sqrt{2} \sqrt{\sinh(t)^{-a} \sinh(t)^3} e^{(-4ax+2x)} \right) dx$$

$$n = \left(\frac{1}{2} \sqrt{2} \sqrt{\sinh(t)^{-a} \sinh(t)^3} e^{(-4ax+2x)} \right) dt + \left(-\sqrt{6} \sqrt{2} \sqrt{\sinh(t)^{-a} \sinh(t)^3} e^{(-4ax+2x)} \right) dx$$

$$m = \left(\frac{i \sqrt{2} b \cosh(t) e^{(-2ax+2x)} \sqrt{\sinh(t)}}{2 \sqrt{\sinh(t)^a}} \right) dx + \left(\frac{1}{2} \sqrt{2} \sqrt{\sinh(t)^a} e^{(2ax-6x)} \sqrt{\sinh(t)} \right) dy + \left(\frac{i \sqrt{2} e^{(4x)} \sqrt{\sinh(t)}}{2 \sqrt{\sinh(t)^a}} \right) dz$$

$$\bar{m} = \left(-\frac{i \sqrt{2} b \cosh(t) e^{(-2ax+2x)} \sqrt{\sinh(t)}}{2 \sqrt{\sinh(t)^a}} \right) dx + \left(\frac{1}{2} \sqrt{2} \sqrt{\sinh(t)^a} e^{(2ax-6x)} \sqrt{\sinh(t)} \right) dy + \left(-\frac{i \sqrt{2} e^{(4x)} \sqrt{\sinh(t)}}{2 \sqrt{\sinh(t)^a}} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{2 \sinh(t)^{\frac{3}{2}}} \right) dt + \left(\frac{\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} \right) dx + \left(-\frac{\sqrt{3} b \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(2ax-4x)}}{12 \sinh(t)^{\frac{3}{2}}} \right) dz$$

$$n = \left(-\frac{\sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{2 \sinh(t)^{\frac{3}{2}}} \right) dt + \left(-\frac{\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} \right) dx + \left(\frac{\sqrt{3} b \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(2ax-4x)}}{12 \sinh(t)^{\frac{3}{2}}} \right) dz$$

$$m = \left(\frac{\sqrt{2} \sinh(t)^{-\frac{1}{2}a} e^{(-2ax+6x)}}{2 \sqrt{\sinh(t)}} \right) dy + \left(\frac{i \sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(-4x)}}{2 \sqrt{\sinh(t)}} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} \sinh(t)^{-\frac{1}{2}a} e^{(-2ax+6x)}}{2 \sqrt{\sinh(t)}} \right) dy + \left(-\frac{i \sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(-4x)}}{2 \sqrt{\sinh(t)}} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{2 \sinh(t)^{\frac{3}{2}}} \partial_t + \frac{\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} \partial_x - \frac{\sqrt{3} b \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(2ax-4x)}}{12 \sinh(t)^{\frac{3}{2}}} \partial_z$$

$$\Delta = -\frac{\sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{2 \sinh(t)^{\frac{3}{2}}} \partial_t - \frac{\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} \partial_x + \frac{\sqrt{3} b \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(2ax-4x)}}{12 \sinh(t)^{\frac{3}{2}}} \partial_z$$

$$\delta = \frac{\sqrt{2} \sinh(t)^{-\frac{1}{2}a} e^{(-2ax+6x)}}{2\sqrt{\sinh(t)}} \partial_y + \frac{i\sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(-4x)}}{2\sqrt{\sinh(t)}} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2} \sinh(t)^{-\frac{1}{2}a} e^{(-2ax+6x)}}{2\sqrt{\sinh(t)}} \partial_y - \frac{i\sqrt{2} \sinh(t)^{\frac{1}{2}a} e^{(-4x)}}{2\sqrt{\sinh(t)}} \partial_z$$

Spin Coefficients

$$\rho = -\frac{\sqrt{3}a \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} + \frac{\sqrt{2} \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(4ax-2x)}}{4 \sinh(t)^{\frac{5}{2}}} + \frac{\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}}$$

$$\sigma = \frac{\sqrt{2}a \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(4ax-2x)}}{4 \sinh(t)^{\frac{5}{2}}} - \frac{\sqrt{3}a \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} + \frac{5\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{3}a \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} - \frac{\sqrt{2} \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(4ax-2x)}}{4 \sinh(t)^{\frac{5}{2}}} + \frac{\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}}$$

$$\lambda = -\frac{\sqrt{2}a \sinh(t)^{\frac{1}{2}a} \cosh(t) e^{(4ax-2x)}}{4 \sinh(t)^{\frac{5}{2}}} - \frac{\sqrt{3}a \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}} + \frac{5\sqrt{3} \sinh(t)^{\frac{1}{2}a} e^{(4ax-2x)}}{12 \sinh(t)^{\frac{3}{2}}}$$

$$\nu = 0$$

$$\alpha = \frac{i\sqrt{3}b \sinh(t)^{\frac{1}{2}a} e^{(6ax-2x)}}{48 \sinh(t)^{\frac{3}{2}}}$$

$$\beta = -\frac{i\sqrt{3}b \sinh(t)^{\frac{1}{2}a} e^{(6ax-2x)}}{48 \sinh(t)^{\frac{3}{2}}}$$

$$\tau = -\frac{i\sqrt{3}b\sinh(t)^{\frac{1}{2}a}e^{(6ax-2x)}}{24\sinh(t)^{\frac{3}{2}}}$$

$$\pi = \frac{i\sqrt{3}b\sinh(t)^{\frac{1}{2}a}e^{(6ax-2x)}}{24\sinh(t)^{\frac{3}{2}}}$$

$$\epsilon = \frac{\sqrt{2}a\sinh(t)^{\frac{1}{2}a}\cosh(t)e^{(4ax-2x)}}{8\sinh(t)^{\frac{5}{2}}} - \frac{\sqrt{3}a\sinh(t)^{\frac{1}{2}a}e^{(4ax-2x)}}{6\sinh(t)^{\frac{3}{2}}} - \frac{3\sqrt{2}\sinh(t)^{\frac{1}{2}a}\cosh(t)e^{(4ax-2x)}}{8\sinh(t)^{\frac{5}{2}}} + \frac{\sqrt{3}\sinh(t)^{\frac{1}{2}a}e^{(4ax-2x)}}{12\sinh(t)^{\frac{3}{2}}}$$

$$\gamma = -\frac{\sqrt{2}a\sinh(t)^{\frac{1}{2}a}\cosh(t)e^{(4ax-2x)}}{8\sinh(t)^{\frac{5}{2}}} - \frac{\sqrt{3}a\sinh(t)^{\frac{1}{2}a}e^{(4ax-2x)}}{6\sinh(t)^{\frac{3}{2}}} + \frac{3\sqrt{2}\sinh(t)^{\frac{1}{2}a}\cosh(t)e^{(4ax-2x)}}{8\sinh(t)^{\frac{5}{2}}} + \frac{\sqrt{3}\sinh(t)^{\frac{1}{2}a}e^{(4ax-2x)}}{12\sinh(t)^{\frac{3}{2}}}$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & \frac{\sqrt{3}\sqrt{2}a^2\sinh(t)^a\cosh(t)e^{(8ax-4x)}}{12\sinh(t)^4} - \frac{\sqrt{3}\sqrt{2}a\sinh(t)^a\cosh(t)e^{(8ax-4x)}}{6\sinh(t)^4} - \frac{a^2\sinh(t)^ae^{(8ax-4x)}}{3\sinh(t)^3} \\ & + \frac{a\sinh(t)^ae^{(8ax-4x)}}{4\sinh(t)^3} - \frac{a^2\sinh(t)^ae^{(8ax-4x)}}{8\sinh(t)^5} - \frac{a\sinh(t)^ae^{(8ax-4x)}}{4\sinh(t)^5} + \frac{7\sinh(t)^ae^{(8ax-4x)}}{8\sinh(t)^5}\end{aligned}$$

$$\Phi_{01} = -\frac{iab\sinh(t)^ae^{(10ax-4x)}}{24\sinh(t)^3}$$

$$\Phi_{02} = \frac{b^2\sinh(t)^ae^{(12ax-4x)}}{192\sinh(t)^3} - \frac{a^2\sinh(t)^ae^{(8ax-4x)}}{24\sinh(t)^3} + \frac{a\sinh(t)^ae^{(8ax-4x)}}{2\sinh(t)^3} - \frac{5\sinh(t)^ae^{(8ax-4x)}}{24\sinh(t)^3}$$

$$\Phi_{10} = \frac{iab\sinh(t)^ae^{(10ax-4x)}}{24\sinh(t)^3}$$

$$\Phi_{11} = -\frac{b^2 \sinh(t)^a e^{(12ax-4x)}}{128 \sinh(t)^3} - \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{16 \sinh(t)^3} - \frac{a \sinh(t)^a e^{(8ax-4x)}}{12 \sinh(t)^3} - \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{16 \sinh(t)^5} + \frac{5 \sinh(t)^a e^{(8ax-4x)}}{16 \sinh(t)^3} - \frac{a \sinh(t)^a e^{(8ax-4x)}}{8 \sinh(t)^5} + \frac{7 \sinh(t)^a e^{(8ax-4x)}}{16 \sinh(t)^5}$$

$$\Phi_{12} = -\frac{i ab \sinh(t)^a e^{(10ax-4x)}}{24 \sinh(t)^3}$$

$$\Phi_{20} = \frac{b^2 \sinh(t)^a e^{(12ax-4x)}}{192 \sinh(t)^3} - \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{24 \sinh(t)^3} + \frac{a \sinh(t)^a e^{(8ax-4x)}}{2 \sinh(t)^3} - \frac{5 \sinh(t)^a e^{(8ax-4x)}}{24 \sinh(t)^3}$$

$$\Phi_{21} = \frac{i ab \sinh(t)^a e^{(10ax-4x)}}{24 \sinh(t)^3}$$

$$\Phi_{22} = -\frac{\sqrt{3}\sqrt{2}a^2 \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{12 \sinh(t)^4} + \frac{\sqrt{3}\sqrt{2}a \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{6 \sinh(t)^4} - \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{3 \sinh(t)^3} + \frac{a \sinh(t)^a e^{(8ax-4x)}}{4 \sinh(t)^3} - \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{8 \sinh(t)^5} - \frac{a \sinh(t)^a e^{(8ax-4x)}}{4 \sinh(t)^5} + \frac{7 \sinh(t)^a e^{(8ax-4x)}}{8 \sinh(t)^5}$$

$$\Lambda = \frac{b^2 \sinh(t)^a e^{(12ax-4x)}}{1152 \sinh(t)^3} + \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{144 \sinh(t)^3} + \frac{a \sinh(t)^a e^{(8ax-4x)}}{18 \sinh(t)^3} + \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{48 \sinh(t)^5} - \frac{5 \sinh(t)^a e^{(8ax-4x)}}{144 \sinh(t)^3} + \frac{a \sinh(t)^a e^{(8ax-4x)}}{24 \sinh(t)^5} - \frac{7 \sinh(t)^a e^{(8ax-4x)}}{48 \sinh(t)^5}$$

Weyl Tensor Components

$$\Psi_0 = \frac{\sqrt{3}\sqrt{2}a^2 \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{4 \sinh(t)^4} - \frac{11 a^2 \sinh(t)^a \cosh(t)^2 e^{(8ax-4x)}}{24 \sinh(t)^5} - \frac{5 \sqrt{3}\sqrt{2}a \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{12 \sinh(t)^4} + \frac{5 a \sinh(t)^a \cosh(t)^2 e^{(8ax-4x)}}{3 \sinh(t)^5} + \frac{5 \sqrt{3}\sqrt{2} \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{12 \sinh(t)^4} + \frac{5 a^2 \sinh(t)^a e^{(8ax-4x)}}{24 \sinh(t)^5} - \frac{5 \sinh(t)^a \cosh(t)^2 e^{(8ax-4x)}}{8 \sinh(t)^5} - \frac{11 a \sinh(t)^a e^{(8ax-4x)}}{12 \sinh(t)^5} + \frac{5 \sinh(t)^a e^{(8ax-4x)}}{8 \sinh(t)^5}$$

$$\Psi_1 = -\frac{i \sqrt{3}\sqrt{2}ab \sinh(t)^a \cosh(t) e^{(10ax-4x)}}{96 \sinh(t)^4} - \frac{i \sqrt{3}\sqrt{2}b \sinh(t)^a \cosh(t) e^{(10ax-4x)}}{96 \sinh(t)^4} - \frac{i ab \sinh(t)^a e^{(10ax-4x)}}{48 \sinh(t)^3} - \frac{i b \sinh(t)^a e^{(10ax-4x)}}{16 \sinh(t)^3}$$

$$\Psi_2 = -\frac{b^2 \sinh(t)^a e^{(12ax-4x)}}{144 \sinh(t)^3} + \frac{5a^2 \sinh(t)^a e^{(8ax-4x)}}{72 \sinh(t)^3} + \frac{5a \sinh(t)^a e^{(8ax-4x)}}{36 \sinh(t)^3} + \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{12 \sinh(t)^5} - \frac{25 \sinh(t)^a e^{(8ax-4x)}}{72 \sinh(t)^3} - \frac{a \sinh(t)^a e^{(8ax-4x)}}{12 \sinh(t)^5} + \frac{\sinh(t)^a e^{(8ax-4x)}}{6 \sinh(t)^5}$$

$$\Psi_3 = -\frac{i\sqrt{3}\sqrt{2}ab \sinh(t)^a \cosh(t) e^{(10ax-4x)}}{96 \sinh(t)^4} - \frac{i\sqrt{3}\sqrt{2}b \sinh(t)^a \cosh(t) e^{(10ax-4x)}}{96 \sinh(t)^4} + \frac{iab \sinh(t)^a e^{(10ax-4x)}}{48 \sinh(t)^3} + \frac{ib \sinh(t)^a e^{(10ax-4x)}}{16 \sinh(t)^3}$$

$$\begin{aligned} \Psi_4 = & -\frac{\sqrt{3}\sqrt{2}a^2 \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{4 \sinh(t)^4} + \frac{5\sqrt{3}\sqrt{2}a \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{12 \sinh(t)^4} - \frac{11a^2 \sinh(t)^a e^{(8ax-4x)}}{24 \sinh(t)^3} \\ & - \frac{5\sqrt{3}\sqrt{2} \sinh(t)^a \cosh(t) e^{(8ax-4x)}}{12 \sinh(t)^4} + \frac{5a \sinh(t)^a e^{(8ax-4x)}}{3 \sinh(t)^3} - \frac{a^2 \sinh(t)^a e^{(8ax-4x)}}{4 \sinh(t)^5} - \frac{5 \sinh(t)^a e^{(8ax-4x)}}{8 \sinh(t)^3} + \frac{3a \sinh(t)^a e^{(8ax-4x)}}{4 \sinh(t)^5} \end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"